

Leakage-Aware Multi-Horizon Benchmarking of Deterministic PV Forecasts Across Co-Located Technologies

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Comparisons of photovoltaic (PV) forecasts can be distorted by preprocessing fitted outside the training period, windows that cross temporal splits, unequal valid samples, or weak naive references. We conduct a leakage-aware deterministic benchmark on three co-located PV arrays at the DKA Solar Centre in Alice Springs. Four compact direct multi-output implementations receive six hours of causal history, including past Active Power, and produce one 144-step trajectory whose 1, 4, 8, and 12 h prefixes are evaluated. Preprocessing is fitted on Train only, windows are built independently within each temporal split, checkpoints are selected only on Validation, and Test is excluded from fitting, training, and checkpoint selection. Horizon-specific valid origins form the primary sample set, while complete-H144 origins provide sensitivity analysis. Across 24 array–horizon–scope combinations, an inverted-variate Transformer implementation ranks first 12 times and a depthwise convolutional TCN nine times when Last-value Persistence is included. Their arithmetic mean RMSE skills relative to Last-value Persistence are 0.549 and 0.515, respectively. In a separate point-matched supplementary analysis, Daily Persistence beats the best neural result in 22 of 24 combinations. Rankings vary with array, horizon, and daylight masking, and efficiency rankings differ from accuracy rankings. The main contribution is therefore an auditable evaluation and a persistence-centered application finding, not a new architecture or a claim of universal neural superiority.

Keywords: photovoltaic power forecasting, multi-horizon forecasting, data leakage, persistence benchmark, reproducibility, co-located PV arrays

I. INTRODUCTION

Short-term PV power forecasts support scheduling, reserve allocation, storage coordination, and distribution-system operation, but the stochastic motion of clouds makes accuracy strongly dependent on horizon and available information. Reviews of solar forecasting have consequently emphasized that model comparisons must be interpreted together with forecast horizon, temporal resolution, weather regime, and data source.^{1,2} Recent work has applied recurrent neural networks, temporal convolutional networks (TCNs), Transformer variants, and hybrid decompositions to PV prediction.^{3–6} The resulting literature often reports that a proposed architecture outperforms several learned baselines. Such comparisons can nevertheless be difficult to interpret when basic evaluation choices differ.

Three risks are especially important. First, preprocessing can leak future distributional information if imputation, anomaly detection, or scaling is fitted before the temporal split. Second, a nominally chronological split can still leak if sliding windows cross a split boundary or if the same target interval appears in overlap-

ping subsets. Third, repeated inspection of Test during hyperparameter or checkpoint selection turns Test into another validation set. General machine-learning audits have shown that leakage and weak independence between training and evaluation can inflate performance estimates and undermine reproducibility.^{7–10} Time-series forecasting additionally requires a meaningful naive reference: a learned model may rank first among neural networks while offering no skill over persistence.^{11,12}

PV benchmarking faces another confounder. Comparisons across sites mix differences in technology with differences in weather, orientation, capacity, sensors, and data quality. The DKA Solar Centre in Alice Springs provides co-located arrays and shared environmental measurements, making it possible to compare forecasting behavior while holding site-level meteorology substantially constant.^{4,13} This does not create cross-climate evidence, but it provides a controlled application setting for asking whether conclusions are stable across distinct PV module technologies.

This study therefore asks four questions. (RQ1) Under a unified protocol that excludes future information from preprocessing, window construction, and model selection, are model rankings stable? (RQ2) How do rankings change across 1, 4, 8, and 12 h horizons and between full-timeline and daylight evaluation? (RQ3) Do complex neural models consistently beat last-value persistence, and are their accuracy gains commensurate with

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parameter count and inference cost? (RQ4) Under common meteorological forcing at one site, do these conclusions remain consistent across three PV technologies?

The contribution is empirical and methodological rather than architectural:

1. We implement a leakage-aware protocol with a complete 5-min timeline, Train-only preprocessing, split-local window construction, Validation-only model selection, and sample-matched Test evaluation.
2. We compare three co-located PV technologies under common meteorological observations, reducing site and climate confounding while retaining technology-specific targets.
3. We report four prefixes of a common 12 h trajectory, full-timeline and daylight metrics, three-seed variability, persistence skill, and unified computational efficiency.
4. We document a practically important conditional result: learned models outperform Last-value Persistence on average after causal history power and sample matching are enforced, but Daily Persistence remains stronger in 22 of 24 separate comparisons after its valid point mask is applied identically to every method.

We do not propose a new neural architecture, and we make no claim of state-of-the-art performance, cross-site generalization, or deployment readiness.

II. RELATED WORK

A. PV power forecasting models

PV forecasting has progressed from physical and statistical models toward data-driven systems that combine recent power, irradiance, meteorology, and temporal covariates. General surveys document the parallel development of recurrent, convolutional, attention-based, and probabilistic sequence forecasting.^{14–17} The appropriate method depends on whether future exogenous forecasts are available and on whether the target is minutes, hours, or days ahead.^{1,2} Recurrent models based on long short-term memory (LSTM) networks remain common because they capture ordered dependencies.^{3,18} Convolutional sequence models offer parallel computation and long receptive fields, while attention-based architectures can represent cross-variable and long-range relationships.^{19,20}

Modern time-series architectures were generally proposed outside the PV domain. DLinear challenged the assumption that complex Transformers always dominate long-term forecasting.²¹ Informer introduced efficient attention for long sequences,²² while Autoformer

and FEDformer integrated decomposition or frequency-domain structure.^{23,24} PatchTST represents each variable through temporal patches and channel-independent processing.²⁵ iTransformer inverts the tokenization axis to model variates as tokens.²⁶ ModernTCN revisits convolutional designs with large kernels and modern architectural choices.²⁷ These are established methods; their use in the present benchmark is an evaluation choice, not an algorithmic contribution.

Recent PV studies have used attention, decomposition, recurrent learning, and direct multi-step output.^{3–6} Huang *et al.* evaluated MANODE on three Alice Springs PV series, illustrating both direct relevance of this public facility and the emphasis placed by prior work on architectural novelty.²⁸ We ask a different question: whether compact implementations retain apparent advantages after causal power history, horizon-specific eligibility, and same-sample persistence are enforced. The diversity of published datasets and protocols makes numerical gains difficult to transfer, and our purpose is not to reproduce full official implementations or introduce another architecture.

B. Multi-horizon forecasting and persistence benchmarks

Recursive multi-step forecasting feeds predicted values back into the model and can accumulate error. Direct approaches instead predict each future step without recursion, either with separate models or a joint output trajectory. The latter is attractive when evaluation spans many horizons because one trained model supplies all prefixes. General benchmark studies such as M4 illustrate that no single forecasting method dominates across all series and horizons.¹² The same warning applies to PV forecasting: a model that performs well at 12 h may not be best at 1 h, and a metric averaged over night and day can conceal materially different behavior.

Persistence is especially important for solar data. Last-value persistence is strong over short horizons when power changes slowly and becomes weaker as forecast age grows. A daily seasonal reference can be informative at matched clock times but requires lagged observations and may change the eligible point mask. Skill scores relative to a naive method provide a direct answer to whether complexity adds predictive value.¹¹ We therefore use last-value persistence on exactly the same origins and masks as the neural models. Daily persistence is supplementary; within that analysis, however, its lag-valid point mask is applied identically to Daily Persistence, Last-value Persistence, and all neural predictions.

C. Data leakage and reproducibility risks

Forecasting evaluation should preserve the direction of time. Random cross-validation is generally unsuitable when temporal dependence is present, and even blocked

evaluation can be optimistic if preprocessing is fitted globally.^{8–10} Leakage can also arise indirectly through repeated model selection on the reported test period. Kapoor and Narayanan documented reproducibility failures arising from leakage in data science research more broadly.⁷ In PV applications, missingness, night periods, anomaly treatment, and scaler definitions add further opportunities for unintentional information transfer.

Reproducibility requires more than publishing a metric table. The split, target timestamps, masks, inverse transformation, checkpoint rule, and metric definitions must be traceable. Three random seeds do not establish broad statistical significance, but per-seed results at least reveal whether a ranking is driven by one initialization. Computational comparisons similarly require a common device, input shape, warm-up, batch size, and timing boundary.

D. Identified evaluation gap

The literature contains many PV model comparisons and several studies using Alice Springs data,^{4–6} but fewer studies combine all of the following in one controlled design: co-located arrays with different PV technologies, Train-only preprocessing, split-local windows, an identical direct multi-output horizon, a same-sample persistence challenge, full-timeline and daylight reporting, seed variability, and unified inference efficiency. We describe this as a narrow evaluation gap rather than a claim of first use of any component. The benchmark is intended to show how those choices alter conclusions about established models.

III. DATA AND EXPERIMENTAL PROTOCOL

A. Alice Springs co-located PV systems

The data originate from the DKA Solar Centre in Alice Springs, Australia. Three systems are used: Site 17 with Sanyo HIT-210NKHE5 hybrid-silicon modules (rated DC array capacity 6.30 kW), Site 25 with Hanwha Solar HSL 60S polycrystalline-silicon modules (5.83 kW DC), and Site 38 with Q CELLS Q.PEAK-G4.1 monocrystalline-silicon modules (5.90 kW DC). The official AC capacity is unavailable; therefore the DC ratings are not used to normalize AC power. The target field is Active Power in kW AC according to the DKASC data glossary. The systems share six project weather inputs: ambient temperature, relative humidity, global and diffuse horizontal radiation, and global and diffuse tilted radiation. Official metadata support the technology labels, but latitude and longitude were not available in the final evidence and are not inferred from filenames.

Table I summarizes the temporal protocol. All systems use the same date boundaries, but valid window counts

differ because targets and input features contain system-specific missing intervals. The complete 5-min timeline is retained, including nighttime timestamps.

The use of co-located arrays is a control strategy rather than a claim that the targets are interchangeable. Each array has its own Active Power target, performance ratio, missingness pattern, and Train-fitted target transformation. By contrast, the six weather fields are identical in the clean project data for aligned timestamps. This separation prevents the same weather observation from being counted as three independent meteorological sources while preserving technology-specific response. It also limits the scientific question: the experiment compares systems under common forcing, not generalization to an unseen site.

The available interval is shorter than many operational PV archives, but the five-minute cadence produces thousands of valid forecast origins per split. Adjacent origins share most of their lookback and target intervals and are not independent experimental units. We therefore treat origins as repeated operational forecast instances and random seeds as optimization replications. This is one reason that the study emphasizes descriptive effect sizes and avoids a nominal significance test whose independence assumptions would be difficult to defend.

B. Temporal alignment and data preprocessing

Raw timestamps are reindexed to a regular 5-min axis. Nighttime is not removed, because deletion would change temporal spacing and can make window construction ambiguous. Missing timestamp rows are retained on that coordinate, represented by explicit feature masks, and filled only by Train-fitted imputation. K-nearest-neighbor imputation, Isolation Forest fitting, feature scaling, and target scaling are fitted using Train only. The fitted transformations are then applied without refitting to Validation and Test. KNN imputation and isolation-based anomaly detection are established tools;^{29,30} here they are protocol components rather than methodological contributions.

Each model receives 72 past samples (6 h) and outputs 144 future samples (12 h). The seven original numerical fields are, in fixed order, array-specific Performance Ratio followed by ambient temperature, relative humidity, global horizontal radiation, diffuse horizontal radiation, global tilted radiation, and diffuse tilted radiation. The audited predecessor tensor had 15 channels: these seven values, seven missing-value indicators, and one Isolation Forest indicator. It contained neither cyclical time features nor historical Active Power. The corrected protocol adds past Active Power and its missing-value indicator, yielding 17 channels (eight causal numerical fields, eight masks, and one anomaly indicator). Active Power after the forecast origin is exclusively a target. The power history imputer and scaler are fitted only on Train, using the same discipline as the other transformations.

TABLE I. Dataset and corrected primary Test sample counts. Capacities are official DC array ratings and are not used for AC-power normalization. Counts are common to all neural models and Last-value Persistence; H12–H144 use horizon-specific target validity.

Array (site)	Module technology	Rated DC	Target	H12 origins	H48 origins	H96 origins	H144 origins
Sanyo (17)	HIT hybrid silicon	6.30 kW	Active Power (kW AC)	6,589	5,905	5,024	4,160
Hanwha (25)	Polycrystalline silicon	5.83 kW	Active Power (kW AC)	6,625	6,049	5,281	4,521
Qcells (38)	Monocrystalline silicon	5.90 kW	Active Power (kW AC)	6,463	5,491	4,227	2,996
Train: 2018-04-01–2018-07-15; Validation: 2018-07-16–2018-08-07; Test: 2018-08-08–2018-08-31 (ACST).							

The complete timeline makes missingness visible. Reindexing does not assert that an absent record was observed; it creates a regular coordinate system on which missing indicators and Train-fitted transformations can operate. If night records or missing intervals were silently removed, two adjacent rows could be more than five minutes apart although a model treats them as consecutive samples. The clean protocol instead requires temporal continuity and does not use interpolation to bridge a split boundary.

Preprocessing state is immutable after Train fitting. Validation and Test do not update neighbor sets, anomaly thresholds, feature extrema, or target extrema. This avoids transductive evaluation in which the marginal distribution of future data informs the representation. The same principle applies to inverse transformation: predictions and labels are compared only after one common return to the physical target unit.

C. Leakage-free split and window construction

Figure 1 shows the evaluation flow. Windows are generated separately inside each split; no input or target interval crosses a boundary. Validation loss accumulates squared errors and valid-target counts over the full split before forming one global masked MSE. This replaces equal averaging of batch means, which weights a small final batch incorrectly. Validation alone selects the checkpoint and triggers early stopping, and Test is not passed to the training function. Test was excluded from preprocessing fitting, training, and checkpoint selection. The three seeds use identical labels, forecast-origin timestamps, and masks within each array.

All models are trained and their checkpoints selected using complete H144 Train and Validation windows. They emit one H144 trajectory. Horizon-specific eligibility is introduced only for the primary Test evaluation: L72 history and only the reported H12, H48, H96, or H144 target prefix must be valid. This avoids discarding a valid one-hour forecast merely because a later portion of the 12-hour target is missing. Every model and Last-value Persistence uses identical origins, labels, and masks within a comparison. As sensitivity analysis, we also evaluate all prefixes on complete-H144 Test origins; those results are secondary because the restricted subset systematically lowers errors.

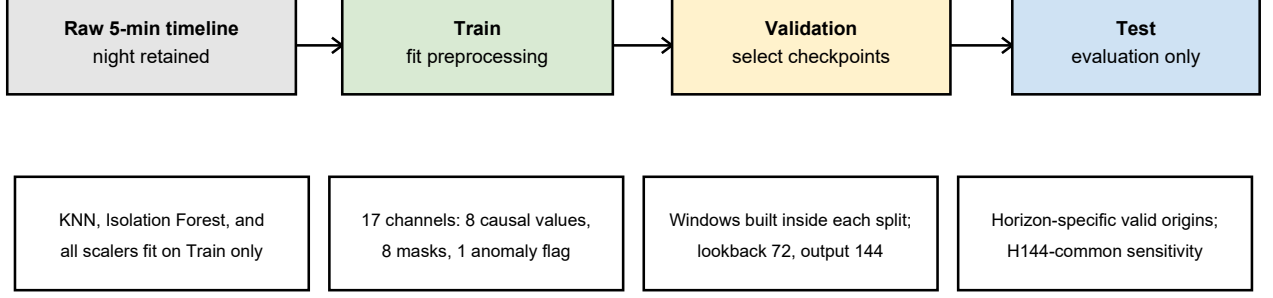
Artifact-level checks confirmed 36 completed runs: four neural models, three arrays, and three seeds. Each run maps to an H144 prediction, label array, forecast-origin timestamps, mask, configuration, and selected checkpoint. Within an array, labels, origins, and masks are elementwise identical across all models and seeds, and all numeric arrays are finite. The selected checkpoints reproduce the saved predictions under no-gradient inference. Metrics in this manuscript were recomputed from those artifacts in the final evidence audit rather than copied from earlier summary tables; this corrected evidence supersedes preliminary project tables.

The daylight mask is defined after prediction for descriptive evaluation: a target point is daylight when its true power exceeds 1% of the corresponding array’s Train maximum. Because this uses the target, it is not an input, routing variable, or deployable daylight detector. It merely removes near-zero target points when computing a second set of metrics.

D. Forecasting models

Four neural models are compared using descriptive names that match the code. The *Discrete recurrent decoder* combines a causal convolutional representation, a compact Transformer block, and recurrent trajectory decoding. The *Inverted-variate Transformer* tokenizes variables and applies one encoder layer. The *Joint-patch Transformer* embeds overlapping temporal patches jointly, and the *Depthwise convolutional TCN* uses four depthwise convolutional blocks. Their broad design ideas are inspired by recurrent/TCN, iTransformer, PatchTST, and ModernTCN literature, respectively,^{19,25–27} but these compact project implementations are not claimed to be complete official reproductions. Every model maps the same 72×17 input to an H144 trajectory and uses seeds 42, 43, and 44. Architecture-specific parameter counts are reported rather than artificially equalized.

The shared optimization configuration is AdamW³¹ with learning rate 0.001, weight decay 10^{-5} , batch size 256, a maximum of 25 epochs, and early-stopping patience 5. The output checkpoint minimizes Validation loss. These settings were fixed before Test evaluation; Test does not select architecture, seed, epoch, or horizon.



Past Active Power is causal input; future Active Power is target only.

Validation global MSE = total SSE / valid targets; Test is excluded from fitting, training, and checkpoint selection.

Leakage-aware and sample-matched temporal evaluation protocol

FIG. 1. Corrected leakage-aware protocol. Preprocessing is fitted exclusively on Train; windows are built independently inside each split; Validation selects checkpoints; and Test is excluded from preprocessing fitting, training, and checkpoint selection.

E. Persistence references

Last-value persistence repeats the most recent observed PV power at the forecast origin across all 144 future steps. It uses the exact neural-model origins, labels, and masks and therefore provides the main naive reference. Daily persistence takes the value at each target timestamp from 24 h earlier (288 five-minute samples), without interpolation. Its actual finite-lag point mask is intersected with the horizon-specific label and daylight masks, and that identical point mask is then applied to Daily Persistence, Last-value Persistence, and all four neural implementations. This gives a fair supplementary comparison, although its counts differ from the primary Last-value analysis and it is therefore not inserted into the primary rank table.

F. Evaluation metrics

For n valid target points with observations y_i and predictions $\hat{y}_i$, RMSE, MAE, bias, and R^2 are

$$\text{RMSE} = \sqrt{n^{-1} \sum_i (y_i - \hat{y}_i)^2}, \quad (1)$$

$$\text{MAE} = n^{-1} \sum_i |y_i - \hat{y}_i|, \quad (2)$$

$$\text{Bias} = n^{-1} \sum_i (\hat{y}_i - y_i), \quad (3)$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \quad (4)$$

Negative R^2 values are retained. Since official AC capacities are unknown, AC RMSE is not divided by the DC array rating. Instead, the primary normalized measure is

$$\text{range-nRMSE} = \frac{\text{RMSE}}{y_{\max, \text{Train}} - y_{\min, \text{Train}}}. \quad (5)$$

The denominator is fixed from Train for each array. Skill relative to last-value persistence is $1 - \text{RMSE}_{\text{model}} / \text{RMSE}_{\text{pers}}$ (and analogously for MAE); positive values indicate improvement. Neural metrics are reported as mean and sample standard deviation across three seeds. We do not interpret the three seeds as an independent-sample significance test.

G. Computational efficiency measurement

Inference was measured on one NVIDIA GeForce RTX 3060 Laptop GPU under Windows and PyTorch using float32. All models used input shape $[B, 72, 17]$ in evaluation mode. Batch-one latency used warm-up followed by synchronized repetitions; throughput used batch 256. Timing excludes model loading, disk I/O, data loading, and host-to-device transfer. These measurements characterize this hardware and software stack only.

H. Evidence construction and reproducibility

The corrected evidence is stored in long format, with one row per array, model, seed, analysis, horizon, scope, statistic, and metric. Each row records physical units,

forecast-origin and valid-target counts, the normalization definition, and artifact provenance. This format separates calculation from manuscript layout: plots and summary tables can be regenerated without transcribing values from a PDF or spreadsheet. Persistence rows record their causal construction rule instead of a checkpoint.

Three consistency levels were checked. At the run level, every expected model-array-seed combination had a completed configuration, H144 prediction, and checkpoint. At the array level, labels, timestamps, and masks were compared elementwise across models and seeds. At the metric level, every H12, H48, H96, and H144 result was recomputed from the corresponding prefix of one H144 artifact. Means, sample standard deviations, and neural ranks were then derived from per-seed rows. The 36 run count refers only to trained neural runs; persistence is deterministic and adds no training run.

Figure generation reads those evidence files directly from their audited project location. The manuscript directory intentionally does not contain a copied metric table, because a duplicate would become a competing source of truth. The Python figure script asserts the expected evidence row count, required model set, and horizon set before drawing vector PDFs. This does not replace independent replication, but it reduces transcription and selective-plotting risk within the present study.

The protocol also distinguishes reproducibility from generalizability. A reader can reproduce the transformation and metric logic on the released data and code, but the Test period remains one temporal sample from one site. Agreement between three seeds addresses optimization variability, not climate variability. Claims in the following sections are therefore framed as observations for the audited Alice Springs task.

IV. RESULTS

A. Overall multi-horizon accuracy

Table II reports primary Train-range nRMSE using horizon-specific valid origins. The Inverted-variate Transformer has the smallest absolute macro error and ranks first in 12 of 24 array-horizon-scope combinations when Last-value Persistence is included. The Depthwise convolutional TCN wins nine, the Joint-patch Transformer two, and the Discrete recurrent decoder one. Last-value Persistence wins none, but it still beats the latter two implementations for Qcells H12.

Physical RMSE supports the normalized comparison. At full-timeline H144, the best learned values are 0.776 kW (Sanyo, Inverted-variate), 0.743 kW (Hanwha, Depthwise TCN), and 0.789 kW (Qcells, Depthwise TCN), compared with Last-value Persistence at 2.729, 2.561, and 2.418 kW. At Qcells H12, the corrected common-sample Persistence RMSE is 0.471 kW, not the previously reported 0.004 kW; Inverted-variate and Depthwise TCN achieve 0.327 and 0.416 kW, whereas

the Discrete recurrent and Joint-patch implementations remain worse.

The three arrays show both common structure and meaningful differences. Inverted-variate leads most Qcells prefixes and Sanyo H96/H144, whereas Depthwise TCN leads most Hanwha settings and short Sanyo daylight. Joint-patch wins both Sanyo H48 scopes, and Discrete recurrent wins Sanyo H12 full timeline. This conditional pattern is more defensible than assigning one architecture a universal Alice Springs advantage.

These values illustrate why physical and normalized errors should appear together. The arrays have similar DC ratings but lack an official common AC-capacity normalization. Physical RMSE is interpretable within each target, whereas Train-range nRMSE supports cautious cross-array comparison using a denominator known at training time. Neither measure licenses comparison with a different site whose instrumentation, clipping, orientation, and target definition may differ.

B. Full-timeline versus daylight evaluation

Figure 2 shows that daylight filtering changes scale and sometimes ordering. Removing near-zero nighttime target points increases normalized errors because retained targets are more variable. For the best H144 implementation, Sanyo rises from 0.128 full timeline to 0.175 daylight, Hanwha from 0.128 to 0.178, and Qcells from 0.131 to 0.191. The change is not an array-specific anomaly.

Rank changes are also visible. Sanyo H12 switches from Discrete recurrent in the full timeline to Depthwise TCN in daylight, while Hanwha H48 switches from Inverted-variate to a near-tie with Depthwise TCN. These differences show why a single aggregate score is insufficient. Daylight is not a deployable selection rule: its mask uses true target power and is only an analysis scope.

C. Cross-technology ranking stability

The heat map in Fig. 3 makes conditional ranking explicit and includes Last-value Persistence rather than separating it after model selection. Inverted-variate has the best mean rank (1.875) but does not dominate all technology-horizon pairs. Depthwise TCN is close (2.167), followed by Joint-patch (2.458), Discrete recurrent (3.667), and Last-value Persistence (4.833). Qcells H12 remains difficult for two compact implementations, yet Persistence is no longer an almost exact or overall-winning reference after causal input and sample matching are corrected.

Table III summarizes win count and average rank. The win count totals 24 combinations (three arrays, four horizons, and two scopes). It is descriptive and does not constitute a significance test.

TABLE II. Primary Test Train-range nRMSE on horizon-specific valid origins. Neural entries are three-seed means (sample SD); Last-value Persistence is deterministic. Bold marks the lowest value. Daily Persistence is evaluated separately with its lag-valid point mask applied identically to every method.

Array	Scope	Horizon	Last persistence	Discrete recurrent	Inverted-variate	Joint-patch	Depthwise TCN
Sanyo	Full	H12	0.0766	0.0342	0.0423	0.0419	0.0365
		H48	0.2376	0.1077	0.1059	0.1016	0.1077
		H96	0.3801	0.1527	0.1363	0.1444	0.1489
		H144	0.4506	0.1433	0.1280	0.1309	0.1348
	Daylight	H12	0.1101	0.0492	0.0524	0.0538	0.0480
		H48	0.3342	0.1583	0.1442	0.1426	0.1494
		H96	0.5205	0.2267	0.1871	0.2063	0.2094
		H144	0.6056	0.2136	0.1745	0.1864	0.1886
	Hanwha	Full	H12	0.0745	0.0366	0.0444	0.0301
		H48	0.2320	0.1025	0.0986	0.1005	0.1007
		H96	0.3731	0.1471	0.1479	0.1388	0.1379
		H144	0.4394	0.1348	0.1382	0.1280	0.1275
		Daylight	H12	0.1078	0.0525	0.0473	0.0396
		H48	0.3223	0.1503	0.1401	0.1417	0.1415
Qcells	Full	H12	0.0780	0.1529	0.0542	0.0882	0.0689
		H48	0.2166	0.1961	0.1338	0.1633	0.1820
		H96	0.3403	0.2078	0.1883	0.1980	0.2252
		H144	0.4005	0.1505	0.1397	0.1408	0.1307
	Daylight	H12	0.1129	0.2223	0.0756	0.1240	0.0992
		H48	0.3156	0.2857	0.1922	0.2352	0.2644
		H96	0.5038	0.3060	0.2729	0.2880	0.3298
		H144	0.6068	0.2251	0.2024	0.2055	0.1909

TABLE III. Primary ranking and Persistence-skill summary over 24 array–horizon–scope combinations. Skill is the arithmetic mean of per-combination RMSE ratios, not an absolute-error average.

Model	RMSE wins	Average rank	Mean skill
Inverted-variate Transformer	12	1.875	0.549
Depthwise convolutional TCN	9	2.167	0.515
Joint-patch Transformer	2	2.458	0.493
Discrete recurrent decoder	1	3.667	0.399
Last-value Persistence	0	4.833	0 (reference)

D. Comparison against persistence

Persistence produces the central qualification. Relative to Last-value Persistence, Inverted-variate and Depthwise TCN have positive RMSE skill in all 24 primary combinations, while Joint-patch and Discrete recurrent are positive in 22. Their arithmetic mean skills are 0.549, 0.515, 0.493, and 0.399. These ratio-based means must not be conflated with mean absolute error or macro Train-range nRMSE, which instead ranks Inverted-variate (0.135), Joint-patch (0.143), Depthwise TCN (0.144), and Discrete recurrent (0.161).

The Qcells H12 correction illustrates why sample counts are part of the result. The old evidence compared 3,019 neural origins with 2,996 Persistence origins and contained only 42 daylight target points. The corrected common set contains 6,463 origins, 77,556 full-timeline targets, and 36,504 daylight targets (47.1

Skill remains a ratio and should be presented with physical RMSE, range-nRMSE, and win counts. A high arithmetic mean of ratios does not mean the same thing as a low arithmetic mean error. We therefore keep these summaries separate throughout the manuscript.

Daily Persistence is the more demanding supplementary reference. Its 24 h lag changes valid point counts, so it is not included in primary ranks. For each supplementary comparison, we apply the same lag-valid point mask to Daily Persistence, Last-value Persistence, and all neural predictions. Under this matched design, Daily Persistence beats the best neural implementation in 22 of 24 combinations; only Hanwha H12 full timeline (0.185 versus 0.176 kW) and daylight (0.273 versus 0.231 kW) favor the Depthwise TCN. Thus, the corrected evidence supports superiority over Last-value Persistence, not blanket superiority over persistence as a class.

E. Accuracy–efficiency trade-off

Table IV and Fig. 4 compare neural efficiency. Joint-patch has the lowest mean batch-one latency (0.535 ms) and the highest throughput (421,929 samples/s), while Depthwise TCN has the largest parameter count (683,024). The Discrete recurrent decoder is smallest (99,362 parameters) but much slower because it decodes the trajectory recurrently. Parameter count alone does not determine runtime.

Primary Test Train-range nRMSE on horizon-specific valid origins

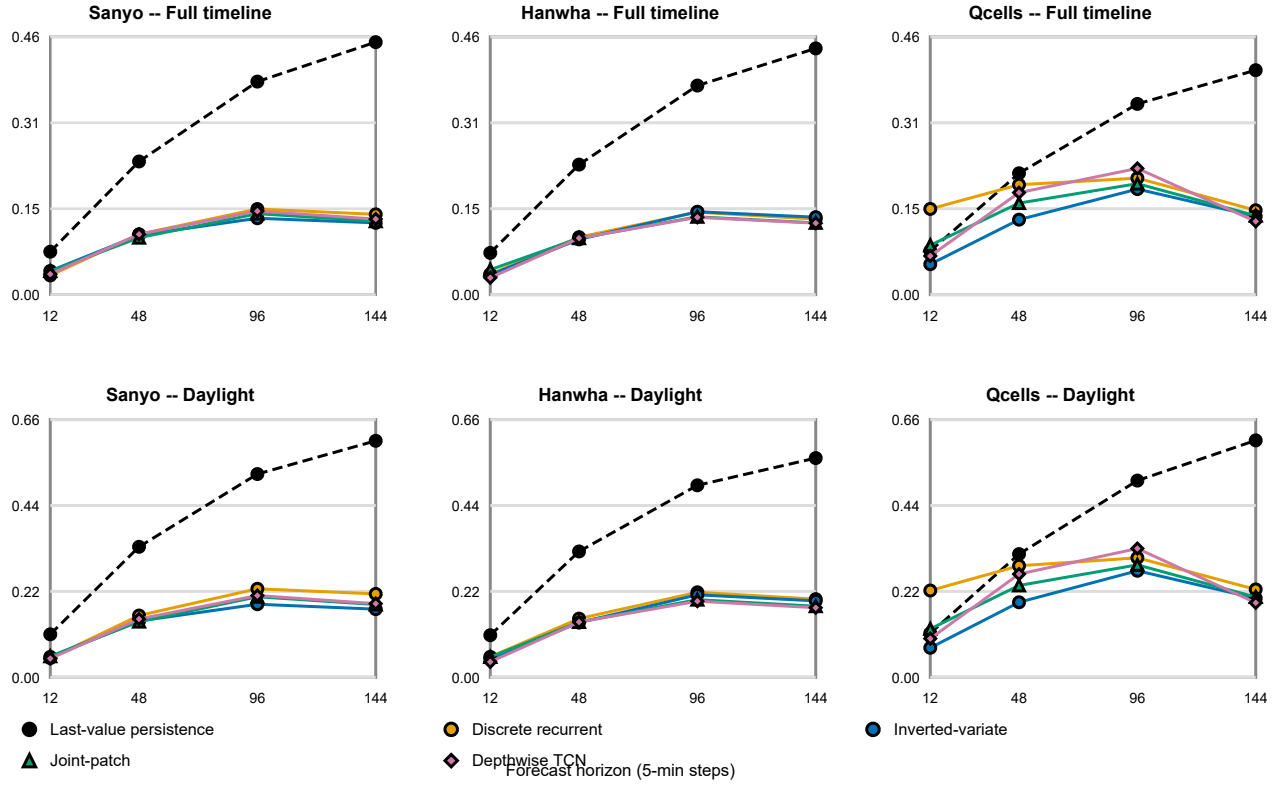


FIG. 2. Primary Test Train-range nRMSE across horizons. Neural curves show three-seed means; Last-value Persistence is deterministic. Within each panel and horizon, all methods use identical horizon-specific origins and masks.

TABLE IV. Unified GPU efficiency measurements. Latency excludes model loading, data loading, disk I/O, and host-to-device transfer. Values are hardware-specific descriptive measurements.

Model	Parameters	Mean latency (ms)	Throughput (samples/s)	Peak GPU memory (MB)
Discrete recurrent	99,362	32.204	1,116	121.12
Inverted-variate	194,960	0.558	377,131	27.31
Joint-patch	148,112	0.535	421,929	26.47
Depthwise TCN	683,024	0.706	197,426	35.27

Primary RMSE rank including Last-value Persistence (1 = best)

Sanyo Full H12	5	1	4	3	2
Sanyo Full H48	5	4	2	1	3
Sanyo Full H96	5	4	1	2	3
Sanyo Full H144	5	4	1	2	3
Sanyo Day H12	5	2	3	4	1
Sanyo Day H48	5	4	2	1	3
Sanyo Day H96	5	4	1	2	3
Sanyo Day H144	5	4	1	2	3
Hamaha Full H12	5	3	2	4	1
Hamaha Full H48	5	4	1	2	3
Hamaha Full H96	5	3	4	2	1
Hamaha Full H144	5	3	4	2	1
Hamaha Day H12	5	4	2	3	1
Hamaha Day H48	5	4	1	3	2
Hamaha Day H96	5	4	3	2	1
Hamaha Day H144	5	4	3	2	1
Qoella Full H12	3	5	1	4	2
Qoella Full H48	5	4	1	2	3
Qoella Full H96	5	3	1	2	4
Qoella Full H144	5	4	2	3	1
Qoella Day H12	3	5	1	4	2
Qoella Day H48	5	4	1	2	3
Qoella Day H96	5	3	1	2	4
Qoella Day H144	5	4	2	3	1

FIG. 3. Primary RMSE ranks across the four compact neural implementations and Last-value Persistence. Rank 1 denotes the lowest error for each array, horizon, and scope.

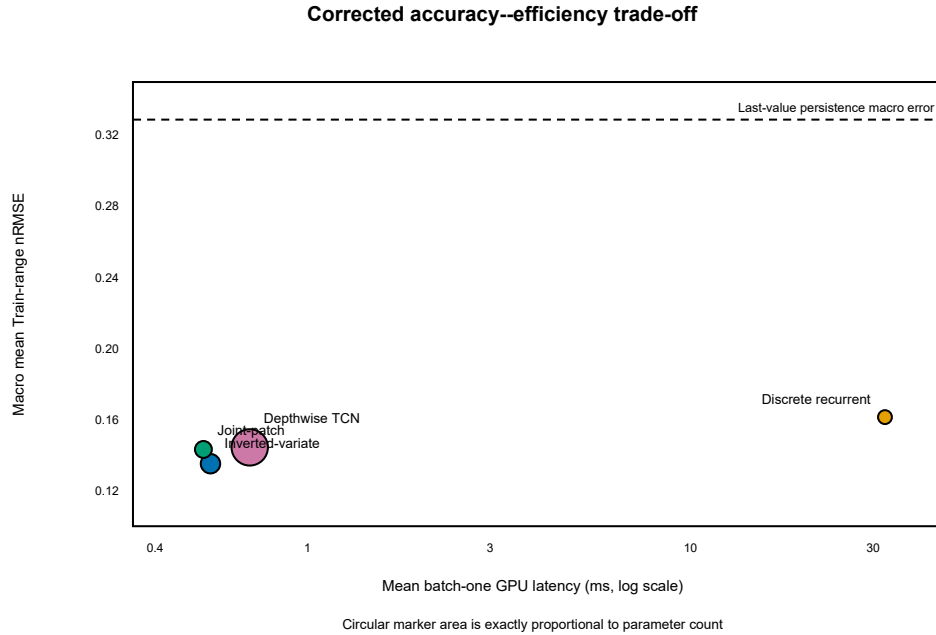


FIG. 4. Corrected accuracy--efficiency trade-off. The vertical coordinate is the unweighted mean of primary Test Train-range nRMSE over 24 array--horizon--scope combinations; circular marker area is exactly proportional to parameter count. The dashed Last-value Persistence reference has no neural inference latency and is not a Pareto point.

No model is best on every efficiency dimension. Inverted-variate has the smallest macro error, Joint-patch the lowest mean latency and highest throughput, and Depthwise TCN the second-best rank with substantially more parameters. The measurements do not estab-

lish performance on embedded controllers, other GPUs, or CPUs.

The throughput ordering is not determined by parameter count because kernels, memory access, and batching affect architectures differently. Joint-patch has both

the largest measured throughput and the smallest mean single-sample latency, whereas the smaller recurrent decoder is much slower. A deployment study would also include data preparation, transfer, batching policy, power consumption, and the target device. The present timing isolates model forward execution so architectural overhead can be compared reproducibly.

V. DISCUSSION

A. What the results mean

The results answer the four research questions. First, ranking is not invariant even when leakage controls and valid origins are fixed: Inverted-variate is strongest in 12 combinations, Depthwise TCN in nine, and two other implementations lead the remainder. Second, horizon and scope alter both scale and ordering. Third, learned models consistently improve on Last-value Persistence only after causal historical power and exact sample matching are enforced; on a separately matched Daily-Persistence mask, they win only the two Hanwha H12 comparisons. Fourth, co-location reduces meteorological confounding but does not produce identical conclusions across PV technologies.

The practical lesson is not to reject neural forecasting. It is to require a same-sample naive reference and report where the learned model adds value. A system operator could reasonably prefer a neural forecast for a 12 h trajectory and persistence for very short stable intervals, but designing such routing is outside this study and would require a new calibration protocol. The current evidence supports benchmark conclusions only.

The correction changes two headline conclusions rather than merely changing decimals. The old claim that a model labeled ModernTCN won 18 of 24 neural comparisons is withdrawn, as is the claim that neural models fail to beat Last-value Persistence on average. The corrected findings distinguish arithmetic mean absolute/range-normalized error from the arithmetic mean of ratio-based skill and place Daily Persistence beside, not behind, the learned-model narrative.

The co-located design further shows that technology labels should not be treated as cosmetic dataset names. Different electrical response, data availability, and target dynamics can alter the practical baseline. A method selected on Sanyo alone would miss the Qcells short-horizon reversal. At the same time, the design cannot attribute this difference solely to cell technology because orientation, balance-of-system behavior, and sensor properties are not experimentally randomized. We use technology as a controlled descriptive factor, not a causal treatment.

B. Why complex models fail to consistently beat persistence

Last-value persistence exploits temporal smoothness without estimating parameters, whereas Daily Persistence additionally exploits strong diurnal recurrence. Supplying past Active Power gives learned models the causal state information needed to improve on the last-value reference. Yet the same models remain weaker than daily recurrence in most supplementary comparisons. This gap suggests that short archival evaluation at one season can reward a stable 24 h pattern more than flexible sequence modeling.

Multi-output losses also encourage conditional-mean trajectories. Under unresolved cloud motion, alternative future ramps may average into a smooth prediction. Earlier error analysis in this project identified high-change daylight windows as the dominant source of deterministic trajectory error. That finding is consistent with an information limitation: historical ground observations identify difficult conditions but do not reveal future cloud direction. We did not include forecast-time future numerical weather prediction, satellite imagery, or sky imagery, so the benchmark cannot determine whether those inputs would overcome the limitation.

Architecture affects this trade-off, but the data do not justify attributing rank to a unique mechanism. The compact Inverted-variate implementation is strongest on average, while the Depthwise TCN is competitive at higher parameter cost. Because these are project implementations inspired by published families rather than complete official reproductions, the evidence compares the code as executed and does not revise claims about iTransformer, PatchTST, or ModernTCN themselves.

The absence of future exogenous inputs is consequential. Historical irradiance and power describe the state at forecast origin, but they cannot uniquely determine cloud motion over the next twelve hours. A deterministic squared-error objective then favors an average of plausible futures, producing smooth trajectories during uncertain ramps. More layers or a different decoder cannot create missing future information. This benchmark therefore sets a boundary for history-only deterministic prediction rather than a ceiling for all PV forecasting systems.

That boundary also explains why exploratory project routes are not converted into manuscript contributions. High-frequency irradiance, event heads, probabilistic intervals, and constrained decoders were investigated elsewhere in the development record, but they addressed different questions or failed their predefined screens. Including them as main evidence would blur the clean benchmark and risk selective reporting. Only the audited deterministic benchmark and persistence references define this paper’s quantitative evidence base.

C. Implications for PV forecasting evaluation

Six reporting practices follow from the evidence. First, preprocessing fit boundaries should be stated as explicitly as train/test dates. Second, window timestamps should be checked so no lookback or target crosses a split. Third, causal target history should not be accidentally omitted from a persistence-challenged model. Fourth, each horizon needs matched origins and explicit counts. Fifth, persistence must use the same labels and masks, with seasonal persistence separated when lag eligibility differs. Sixth, full-timeline and daylight results should be labeled as different estimands.

The same discipline should extend to metadata. DC array capacity and measured AC power are not interchangeable. Dividing AC RMSE by a DC nameplate value may look like a standard capacity-normalized metric, but it mixes quantities affected by inverter capacity and losses. Because official AC capacities remain unknown, we use a Train-only target range and label it explicitly. This is less portable than a verified AC-capacity denominator, yet it is auditable and does not use Test information.

Full-timeline metrics are also not inherently inferior to daylight metrics. They answer different operational questions. A full day includes periods in which zero or near-zero output is correct and may be relevant to scheduling; daylight isolates production-period difficulty. The problem arises when the chosen scope is unstated or daylight is defined from each model’s prediction. Our mask is fixed from true target power and used only after prediction, so every model sees the same evaluation points.

Seed variability should also be retained. Three seeds are modest, but the sample SDs in Table II show that initialization contributes uncertainty. Choosing the best seed would produce optimistic results. Computational cost deserves similar discipline: parameter count, latency, and throughput measure different properties, and timing without a stated device and boundary is not reproducible.

The controlled cross-technology comparison is informative precisely because it is not cross-site. Shared meteorology reduces one source of variation, allowing technology- and target-specific behavior to become visible. At the same time, shared weather and one location limit external validity. Replication across climates and independently instrumented sites would be necessary before claiming generality.

VI. LIMITATIONS

The study has several limitations. The three systems are co-located at one Alice Springs facility and cover April–August 2018; they do not represent geographic or climatic generalization. Although the modules correspond to three documented technologies, three arrays cannot represent all products within those technolo-

gies. Official AC capacities are unknown, so Train-range nRMSE is used instead of capacity-normalized AC error. The denominator is leakage-safe but dataset-dependent. Primary horizon-specific eligibility also means that origin counts decrease with horizon; complete-H144 sensitivity is supplied so this design choice remains visible.

Only three random seeds are available. Their mean and sample SD describe initialization variability but do not support strong inferential claims. The daylight mask uses true future target power and is evaluation-only; it cannot be used operationally at the forecast origin. Test was excluded from preprocessing fitting, training, and checkpoint selection, but the project has inspected its results during iterative evidence correction; it is therefore not described as a pristine one-time external confirmation.

The benchmark uses historical observations but no future exogenous forecast available at issue time. Consequently, it cannot resolve whether operational NWP, satellite cloud motion, or sky images would improve long-horizon ramp prediction. Daily Persistence requires a smaller lag-valid point set than the primary comparison and is therefore supplementary, even though every method is evaluated on exactly that same mask within the supplementary analysis and its strong result is scientifically central. Efficiency measurements are limited to one laptop GPU and omit data transfer and pipeline overhead. Finally, the four compact project implementations are representative rather than exhaustive, are not official full reproductions of their inspiration papers, and none is an original architecture contribution.

VII. CONCLUSION

This leakage-aware benchmark compared four direct H144 neural implementations with persistence on three co-located PV technologies. The corrected protocol supplies causal past power, fits every transformation on Train only, builds split-local windows, selects checkpoints on complete-H144 Validation trajectories, and matches each model to Persistence on horizon-specific Test origins. Inverted-variate wins 12 of 24 primary combinations and Depthwise TCN nine; both have positive mean skill relative to Last-value Persistence. Supplementary Daily Persistence nevertheless beats the best neural result in 22 of 24 comparisons when its lag-valid point mask is applied identically to every method. Daylight evaluation raises error and changes some rankings, while efficiency rankings differ from accuracy rankings.

The principal conclusion is methodological: credible PV forecasting claims require leakage control, horizon- and scope-specific reporting, same-sample persistence skill, seed variability, and transparent efficiency measurement. Under these controls, negative or conditional findings are as important as win counts. The evidence supports an application benchmark and reproducibility contribution, not a new-architecture or cross-climate claim.

DATA AVAILABILITY

The source PV and meteorological observations are available from the DKA Solar Centre data service and associated official metadata pages.¹³ The manuscript reports derived aggregate evidence only; the original data remain subject to the provider's terms. Exact split dates, feature definitions, window counts, and artifact provenance are provided in the accompanying repository.

CODE AVAILABILITY

The leakage-aware evaluation code, evidence tables, figure-generation script, and manuscript source are available in the project repository at <https://github.com/Zhujiangkun-Yohkoh/PV-forecast>. Large checkpoints and prediction arrays are not included in the manuscript branch.

CONFLICT OF INTEREST

The authors have no conflicts to disclose.

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